HOME NETWORK LIVE TRAFFIC ANALYSIS

Authorized Home-Lab Network Security Assessment

Executive summary: An authorized live network assessment was performed on the 192.168.100.0/24 home LAN using Kali Linux, Windows, Nmap and Wireshark. The work established a network baseline and investigated DNS, ARP, TCP, TLS, QUIC and endpoint behavior. No definitive malicious activity was identified in the captured traffic.

1. Objectives

• Place Kali and Windows on the same LAN.

• Discover active hosts and establish service baselines.

• Capture and analyze live network traffic.

• Investigate anomalies and document evidence.

2. Architecture

• Router: 192.168.100.1

• Windows: 192.168.100.8

• Kali: 192.168.100.108/24, eth0, default route via 192.168.100.1

3. Host Discovery

• Nmap host discovery identified 13 active hosts in 192.168.100.0/24.

• Observed vendors included Huawei, Intel, Nintendo and Espressif, with several unknown hosts.

4. Service Enumeration

• Router: DNS/53 and HTTP/80 open; FTP/21, SSH/22, Telnet/23 and 8022 filtered; 1900/tcp reported tcpwrapped.

• Windows: 7070/tcp reported open as ssl/realserver?; exact application not confirmed.

• Kali: no responsive TCP service among the 1,000 ports scanned.

5. Live Traffic and DNS

• The live capture contained 2,592 packets with 0 dropped packets.

• DNS traffic showed Kali using the router as resolver.

• Repeated trycloudflare.com A/AAAA requests returned NXDOMAIN and were recorded as requiring investigation rather than as proof of compromise.

6. TLS, ARP and TCP

• TLS 1.2/1.3 and QUIC traffic were observed.

• A TLS connection to 104.26.13.205 exposed api.ipify.org in handshake data and was not tied to the failed trycloudflare lookups.

• ARP behavior showed normal local resolution with no demonstrated duplicate IP/MAC ownership.

• TCP conversations were dominated by encrypted web traffic; Kali also communicated with the router on TCP/80.

7. Endpoint Analysis

• The unfiltered IPv4 endpoint view contained 20 IPv4 endpoints.

• Kali generated approximately 2 MB of observed traffic.

• 57.144.38.128 was the largest external endpoint by packet count in the capture, with about 1,519 packets and about 2 MB.

8. Security Assessment

• No definitive evidence of malicious activity was identified in the captured traffic.

• Important follow-up items are the unknown LAN devices, Windows TCP/7070, repeated NXDOMAIN trycloudflare.com lookups, and the limited visibility of the current capture point.

9. Recommendations

• Identify unknown devices from the router client list.

• Validate the Windows 7070/tcp service.

• Review and minimize router management exposure.

• Use a dedicated sensor/monitoring point for full LAN visibility.

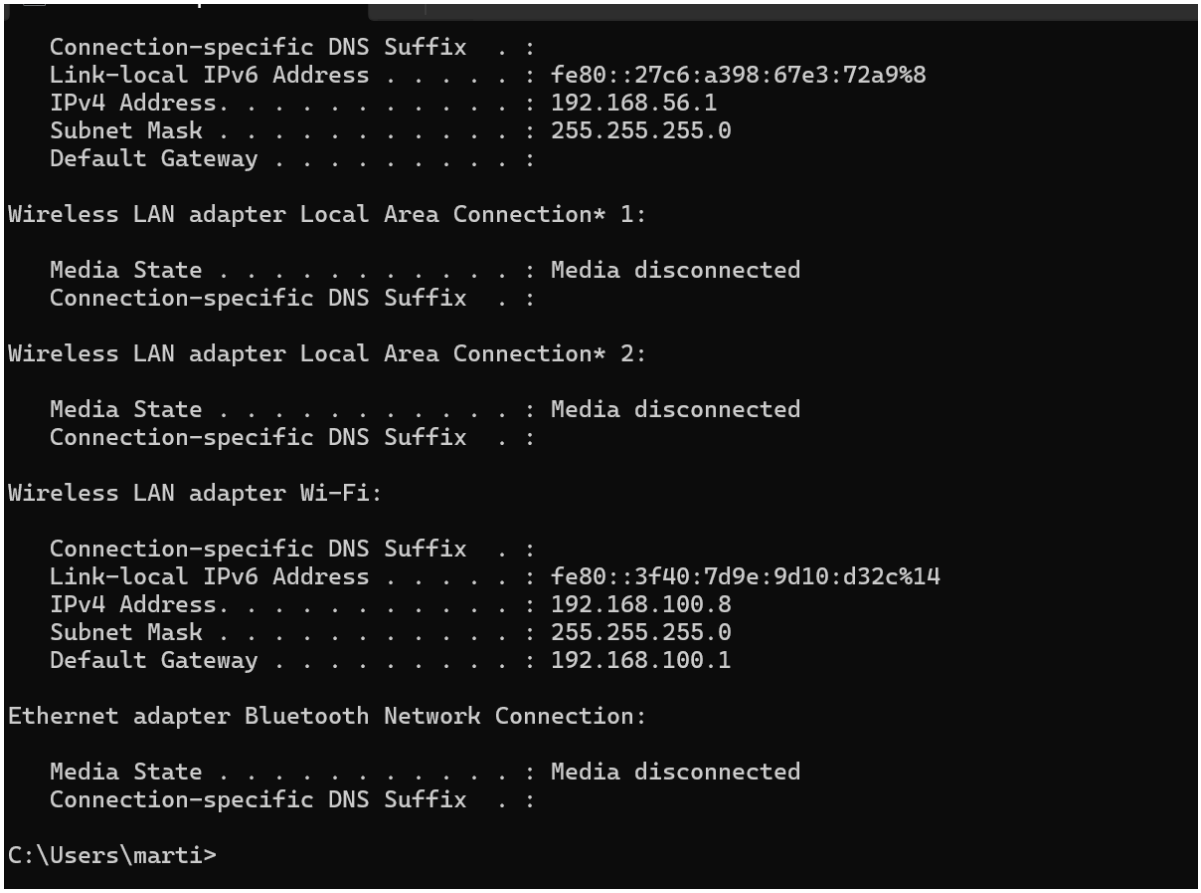
• Maintain PCAP evidence and hash archived artifacts.

10. Conclusion

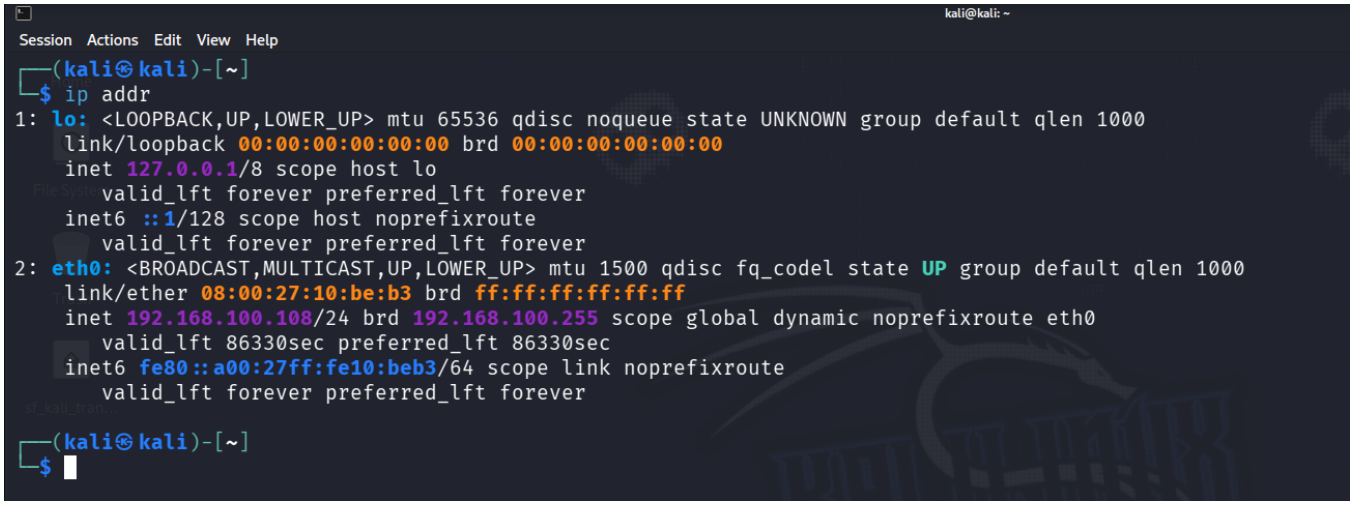
• The project demonstrated a repeatable SOC-style workflow from asset discovery through packet analysis and evidence-based reporting. The results establish a baseline without overstating what a short, single-point capture can prove.

Evidence Appendix

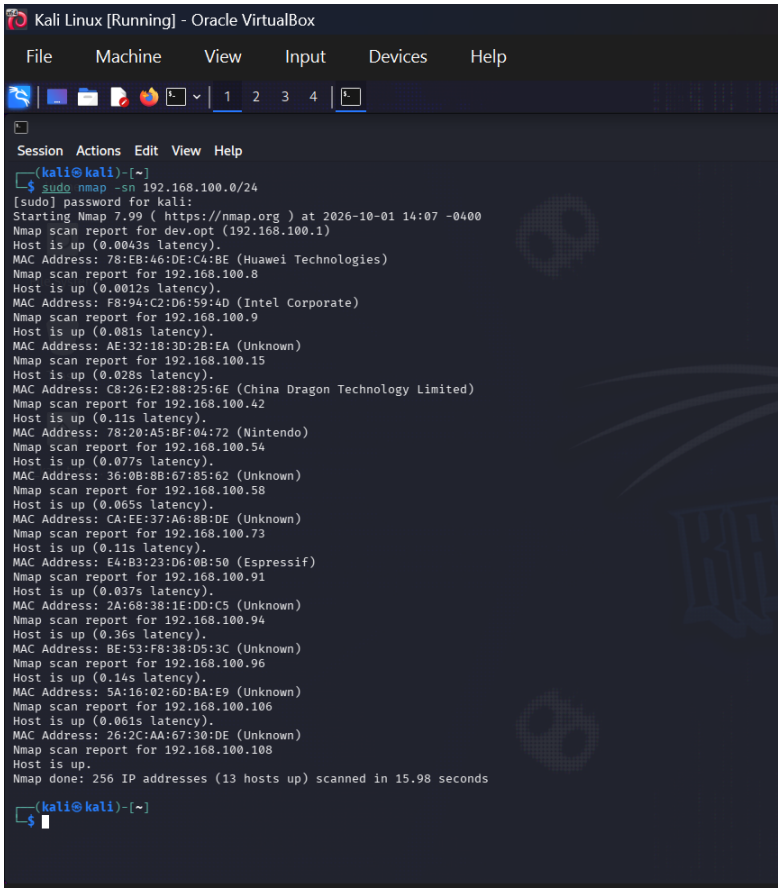
01_windows_ip_configuration.png



02_kali_ip_configuration.png



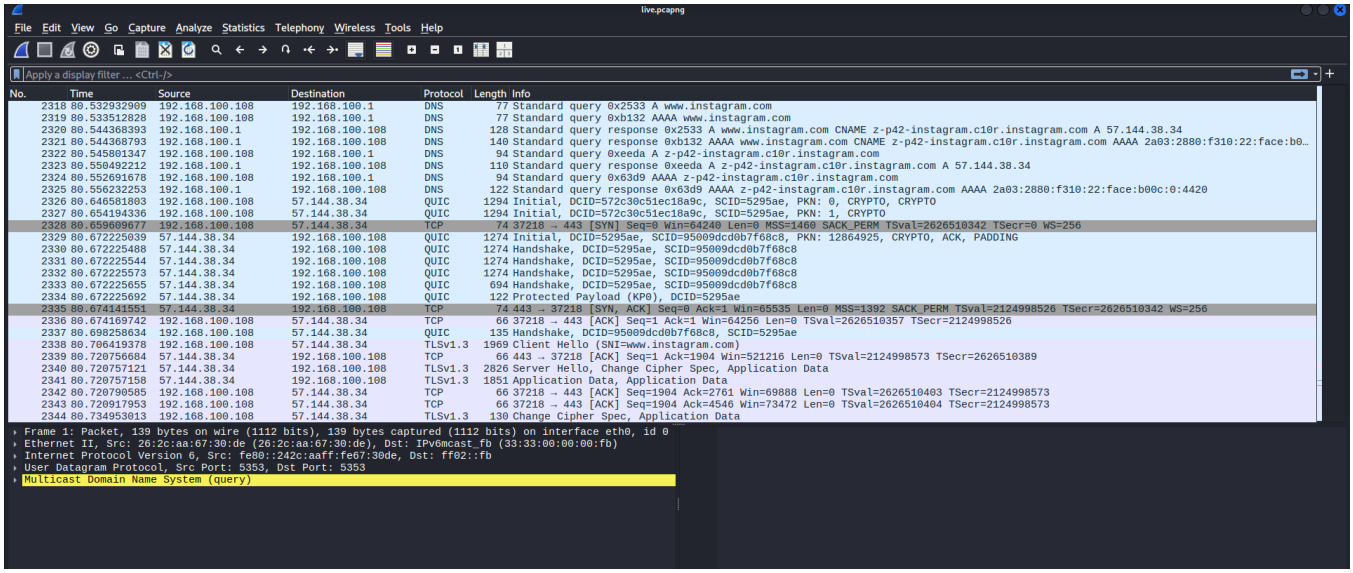
05_network_discovery_nmap.png



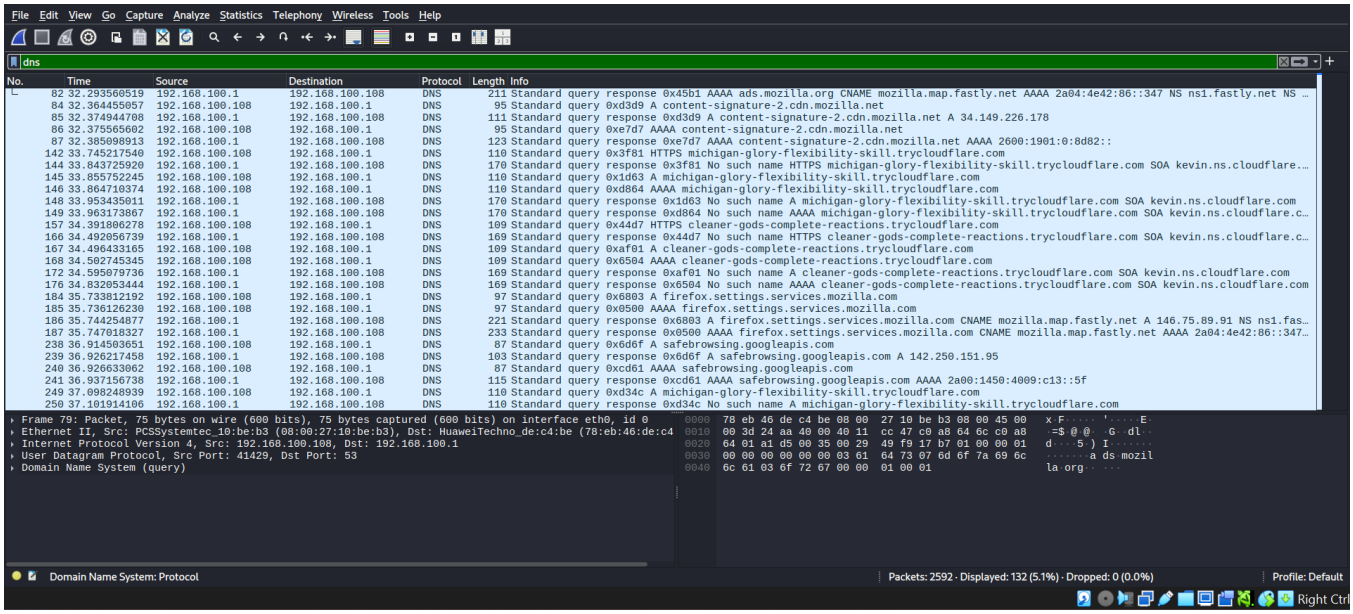
06_router_service_scan.png



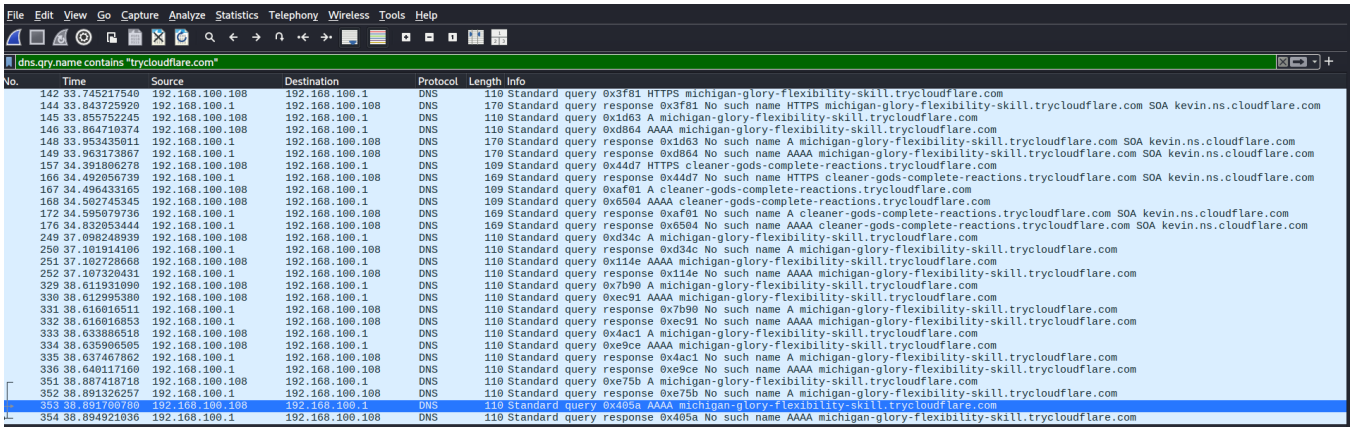
09_live_traffic_capture.png



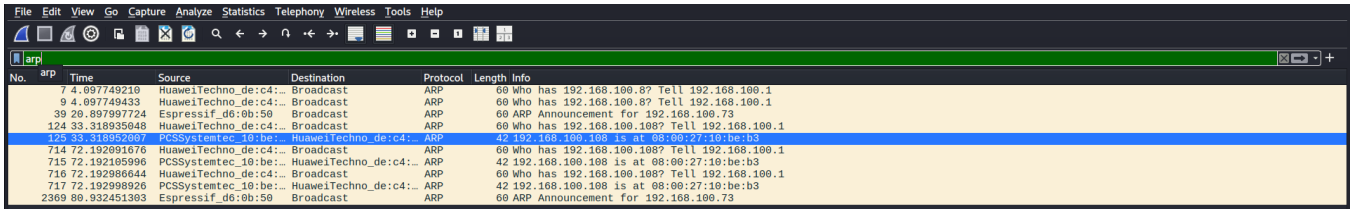
10_dns_traffic_analysis.png



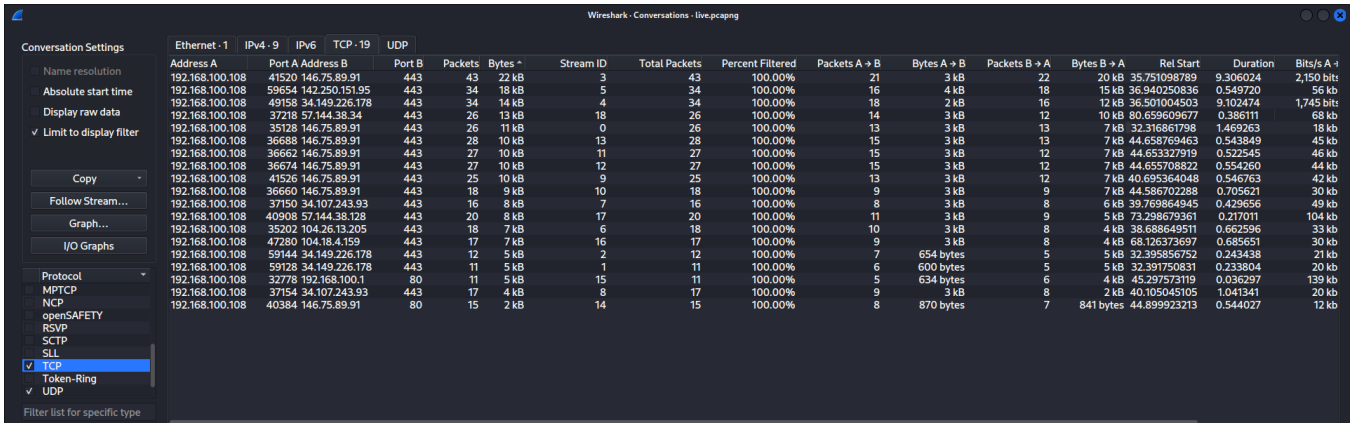
11_trycloudflare_dns_activity.png



13_arp_analysis.png



14_tcp_conversations.png



16_ipv4_endpoints.png

